Class 10 Structural Bioformatics 1

Joshua Khalil A17784122

Table of contents

PDB Statisitics 1
Visualizing HIV Protease Structure 3
Bio3D package for structural bioinformatics 4
Predicting functional motions of a single structure 6

PDB Statisitics

The protein data bank is the main repository of biomolecular structures. lets see what it contains

```
library(readr)
 stats <- read_csv("pdb_stats.csv")
```

```
Rows: 6 Columns: 9
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): Molecular Type
dbl (4): Integrative, Multiple methods, Neutron, Other
num (4): X-ray, EM, NMR, Total

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
 stats
```

```
# A tibble: 6 x 9
  `Molecular Type`    `X-ray`    EM   NMR Integrative `Multiple methods` Neutron
  <chr>                 <dbl> <dbl> <dbl>       <dbl>               <dbl>   <dbl>
1 Protein (only)       178795 21825 12773         343                 226      84
2 Protein/Oligosacch~   10363  3564    34           8                  11       1
3 Protein/NA             9106  6335   287          24                   7       0
4 Nucleic acid (only)    3132   221  1566           3                  15       3
5 Other                   175    25    33           4                   0       0
6 Oligosaccharide (o~      11     0     6           0                   1       0
# i 2 more variables: Other <dbl>, Total <dbl>
```

> Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

The comma in these numbers leads to numbers being read as characters.

```r
n.xray <- sum(stats$`X-ray`)

n.Total <- sum(stats$Total)

n.xray/n.Total
```

```
[1] 0.8095077
```

80.95%

> Q2: What proportion of structures in the PDB are protein?

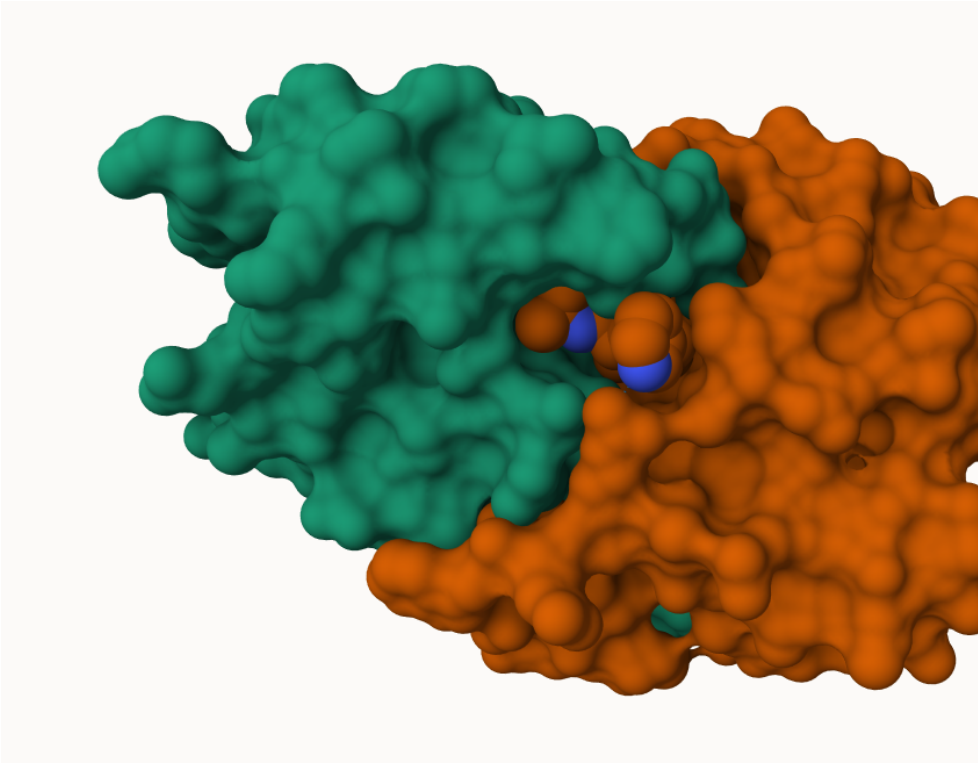
```r
n.Total <- sum(stats$Total)

n.protein <- stats$Total[stats$`Molecular Type` == "Protein (only)"]

n.protein / n.Total
```

```
[1] 0.8596889
```

85.96%

> Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB SKIP?

Visualizing HIV Protease Structure



We can use Molester viewer online.

A new clean image showing the catalytic ASP25 amino acid in both chains of the HIV- PR dimer along with the inhibitor and the important active site water.

> Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

Because the PDB file for this X-ray structure typically only includes the oxygen atom for each water (HOH). The hydrogen atoms are usually not resolved (too hard to see reliably at typical X-ray resolution), so they aren't listed, and you end up with "water oxygen atoms" in the coordinates.

> Q5: There is a critical "conserved" water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

The conserved water is HOH residue 308

Bio3D package for structural bioinformatics

```
library(bio3d)
pdb <- read.pdb("1hsg")
```

```
Note: Accessing on-line PDB file
```

```r
pdb
```

```
 Call:  read.pdb(file = "1hsg")

   Total Models#: 1
     Total Atoms#: 1686,  XYZs#: 5058  Chains#: 2  (values: A B)

     Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 172  (residues: 128)
     Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]

   Protein sequence:
      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
      QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
      ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
      VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

```r
head(pdb$atom)
```

```
  type eleno elety  alt resid chain resno insert      x      y     z o     b
1 ATOM     1     N <NA>   PRO     A     1   <NA> 29.361 39.686 5.862 1 38.10
2 ATOM     2    CA <NA>   PRO     A     1   <NA> 30.307 38.663 5.319 1 40.62
3 ATOM     3     C <NA>   PRO     A     1   <NA> 29.760 38.071 4.022 1 42.64
4 ATOM     4     O <NA>   PRO     A     1   <NA> 28.600 38.302 3.676 1 43.40
5 ATOM     5    CB <NA>   PRO     A     1   <NA> 30.508 37.541 6.342 1 37.87
6 ATOM     6    CG <NA>   PRO     A     1   <NA> 29.296 37.591 7.162 1 38.40
  segid elesy charge
1  <NA>     N   <NA>
2  <NA>     C   <NA>
3  <NA>     C   <NA>
4  <NA>     O   <NA>
5  <NA>     C   <NA>
6  <NA>     C   <NA>
```

```
#library(bio3dview)
#view.pdb(pdb)
```

```
#library(bio3dview)
#Library(NGLVieweR)

#view.pdb(pdb) |>
  #setSpin()

# Select the important ASP 25 residue
#sele <- atom.select(pdb, resno=25)

# and highlight them in spacefill representation
#view.pdb(pdb, cols=c("navy","teal"),
       #  highlight = sele,
        # highlight.style = "spacefill") |>
 # setRock()
```

Predicting functional motions of a single structure

Read an ADK structure from PDB database

```
adk<- read.pdb("6s36")
```

```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

```
adk
```

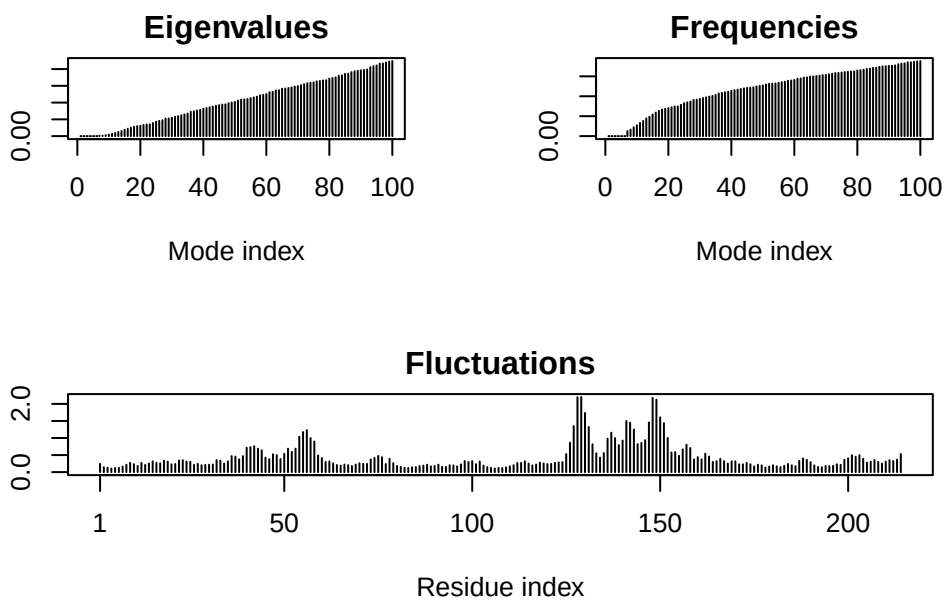
```
 Call:  read.pdb(file = "6s36")

   Total Models#: 1
     Total Atoms#: 1898,  XYZs#: 5694  Chains#: 1  (values: A)

     Protein Atoms#: 1654  (residues/Calpha atoms#: 214)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 244  (residues: 244)
     Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

```
   Protein sequence:
      MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
      DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
      VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
      YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

```r
m<- nma(adk)
```

```
 Building Hessian...        Done in 0.013 seconds.
 Diagonalizing Hessian...   Done in 0.057 seconds.
```

```r
plot(m)
```



Write out our results as we trahectory/ movie of predicted motions:

```r
mktrj(m, file="adk_m7.pdb")
```